Odor-weighted maps of APL synapses

> **QUESTION**
>
> How does each APL synapse in the calyx get an odor response, how are the odor-weighted maps drawn, and does weighting by APL's synapses on claws change how similar two odors look?

INPUTS

- Yang's ORN table `ORN_ovlp16_resp_flyave_merge_dropna.csv` (36 glomeruli × 16 odors, fly means)
- Yang's per-fly ORN pickles `orn_resp_megamat_trialmean_resp.pkl` (9 flies, 17 odors) and `orn_resp_validation2_trialmean_resp.pkl` (5 flies, 22 odors)
- Yang's GH146 PN imaging `GH146_ij_certain-roi_stats.csv` (7 flies, 29 glomeruli, 41 odor presentations, 3 repeats)
- The APL tables, claws and boutons of Reports 01 and 02

OUTPUTS

- `odor_weighted/PN_APL_synapses_*_weighted.csv`, `KC_APL_synapses_on_claws_by_glomerulus.csv` (synapse tables with weights)
- `odor_weighted/*_map_*_posterior_view.png` (the maps)
- `ORN_fly_means_by_glomerulus.csv`, `GH146_mean_by_glomerulus.csv`
- `odor_weighted/odor_similarity_KC_APL_on_claws.csv/.png`

SCRIPTS

- `scripts/odor_meta.py`, `odor_maps.py`, `gh146_maps.py`, `glom_maps.py`, `orn_load.py`, `orn_maps.py`, `claw_glom_maps.py`, `gh146_per_odor_maps.py`, `odor_similarity.py`
- `scripts/reports/r03_odor_maps.py` (this report)

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The maps were made to see where in the calyx APL meets the PNs and claws that a given odor drives. The same glomerulus-to-response step is what feeds odors into the model (Report 05), so it is described here in full. All maps are in both versions of the tables (official ROIs, and with the synapses outside them added; Report 02).

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
g	An antennal-lobe glomerulus (FlyWire names; hemibrain VC3l, VC3m, VC5 renamed VC3, VC5, VM6)	36 in the ORN table, 29 in GH146
o	An odor presentation: odor name and log10 dilution, e.g. "pa @ −3"	
f	A fly (ORN or GH146 imaging)	
$r_{g,o}^{(f)}$	Response of glomerulus g to odor o in fly f: ORN = trial-mean response as given in Yang's pickles; GH146 = peak ΔF/F of one trial	units as in Yang's files; ΔF/F for GH146
$\mathcal{F}_{g,o}$	The flies in which glomerulus g was imaged with odor o	
$\bar{r}_{g,o}$	Fly-mean response (Equation 1)	same units

Symbol	What it stands for	Values, units, range
$g(e)$	Glomerulus of APL synapse e: of its PN (PN↔APL) or of its claw's main PN (KC↔APL on a claw)	
$w_o(e)$	Weight of synapse e for odor o: $\bar{r}_{g(e),o}$	blank if $g(e)$ has no response
$v_{\min}$, $v_{\max}$	Smallest and largest value of the whole mean matrix; the ends of the color scale	ORN −0.476 / 3.503; GH146 −0.158 / 1.962
ρ_{ab}	Similarity of odors a and b in Section 5: Pearson correlation over synapses (or over glomeruli)	−1 to 1

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1 Odor responses per glomerulus

Three response tables from Yang are used. The **ORN table** gives the response of the olfactory receptor neurons of each glomerulus (pebbled imaging) to the 16 odors of the TNT / TNT3c experiment; it is the input to the model. The **per-fly ORN pickles** give the same kind of response per fly, for 39 odors in two panels (megamat: 17 odors at 10^{-3}; validation2: 22 odors at several dilutions). The **GH146 table** gives peak ΔF/F of the projection neurons (GH146 line) in 29 glomeruli, per fly and trial, for 41 odor presentations (24 in a "diagnostics" panel and 17 in megamat). Each is averaged over flies:

$$\bar{r}_{g,o} = \frac{1}{|\mathcal{F}_{g,o}|} \sum_{f \in \mathcal{F}_{g,o}} r_{g,o}^{(f)} \tag{1}$$

ORN: flies where the glomerulus was imaged (missing values skipped). GH146: every fly has 3 repeats, so the mean over flies and repeats equals the mean of per-fly means.

The fly means of the ORN pickles reproduce the ORN table exactly (largest difference 2×10^{-16}), so the table is the same data. For the 16 model odors GH146 has 13: the 12 megamat odors at 10^{-3} and paa at 10^{-5} (diagnostics panel); −aPine, euc and B-myr were not presented.

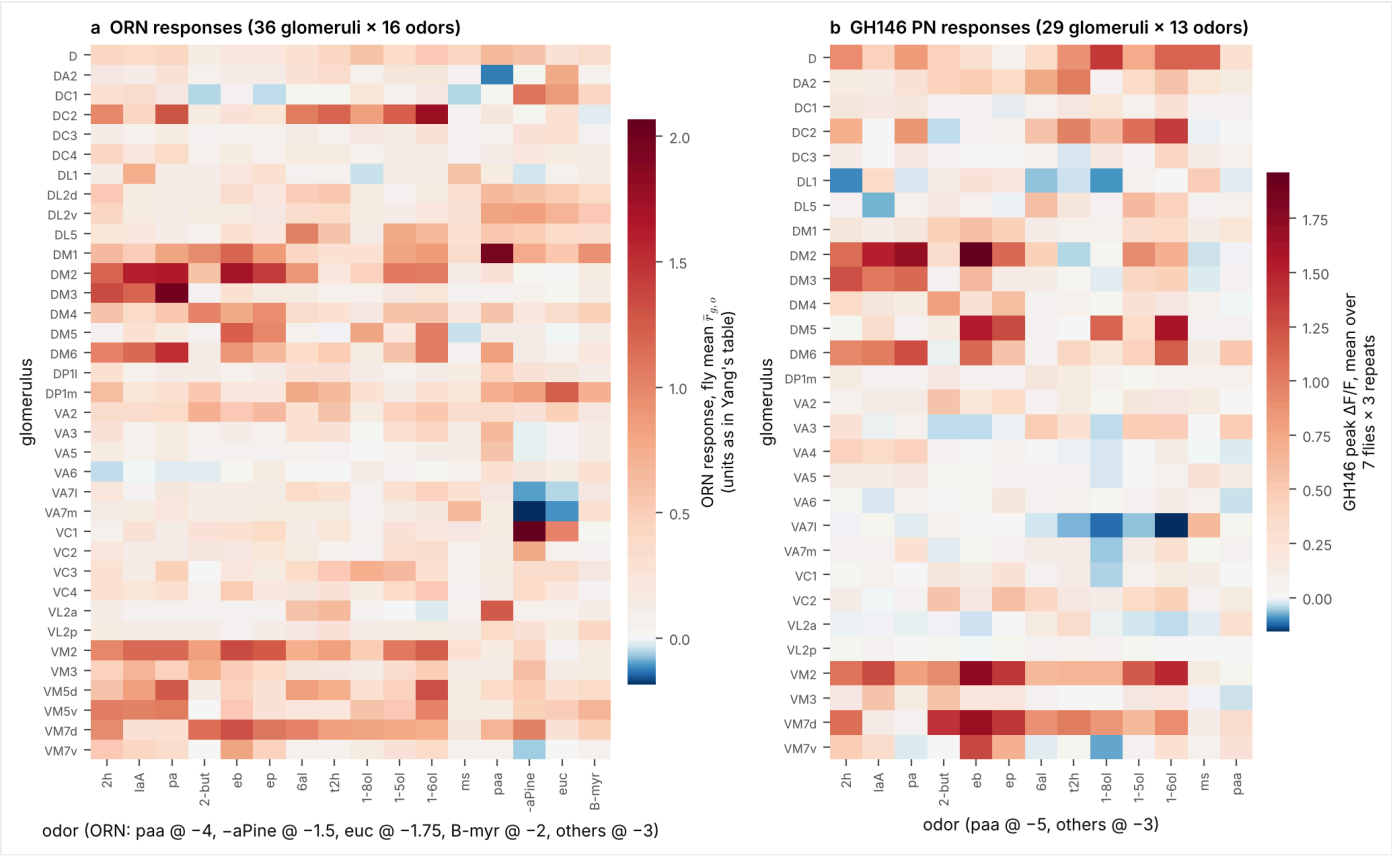


Figure 1. The odor response tables used throughout. Rows: glomeruli; columns: the 16 odors used by the model (GH146 lacks three of them). Color: fly-mean response (Equation 1) on a red–white–blue scale whose white is exactly 0, blue end the smallest value and red end the largest value of each panel's matrix (two-slope scale: negative and positive values are stretched separately, so the colorbar is linear in value on each side of 0). The two panels have their own scales. Odor abbreviations follow Yang's files (pa: pentyl acetate, paa: phenylacetaldehyde, ms: methyl salicylate, eb: ethyl butyrate, ep: ethyl propionate, 2h: 2-heptanone, IaA: isoamyl acetate, 2-but: 2-butanone, 6al: hexanal, t2h: trans-2-hexenal, 1-5ol / 1-6ol / 1-8ol: pentanol / hexanol / octanol, −aPine: (−)-α-pinene, euc: eucalyptol, B-myr: β-myrcene).

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2 Giving each synapse a glomerulus

- **PN ↔ APL synapses.** The PN's cell type names its glomerulus (e.g. `DM1_lPN` → DM1). Only uniglomerular types count. Types that name two glomeruli (VP1d+VP4, VP1m+VP5, VP3+VP1l) count for both; "X+" types (VP2+, VP3+) count for X. Multiglomerular (M_) and untyped PNs get no glomerulus: 1.2–1.4% of PN ↔ APL calyx synapses.
- **KC ↔ APL synapses.** Only synapses **on a claw** (KC-side site within 2 µm of the claw, Report 02) get a glomerulus: that of the claw's main PN. 18–27% of KC → APL and 30–49% of APL → KC calyx synapses are on a claw (lowest in BANC; Report 02, Figure 3).
- **Names.** Hemibrain uses older names for three glomeruli; they are renamed to FlyWire's (VC3l → VC3, VC3m → VC5, VC5 → VM6), which the response tables use.

Each synapse e then carries, for each odor, the response of its glomerulus:

$$w_o(e) \;=\; \bar{r}_{g(e),\,o} \tag{2}$$

Blank (drawn gray) when $g(e)$ is not in the response table, or e has no glomerulus.

Table 2. How many PN ↔ APL calyx synapses can be colored. Both directions and both APLs together, official ROIs. The rest belong to glomeruli that were not imaged (mostly VP thermo- and hygrosensory glomeruli and a few olfactory ones).

Dataset	PN ↔ APL synapses in the calyx	with a glomerulus	glomerulus in the ORN table (36)	glomerulus in GH146 (29)

Dataset	PN ↔ APL synapses in the calyx	with a glomerulus	glomerulus in the ORN table (36)	glomerulus in GH146 (29)
FlyWire 783	10,001	95%	78%	62%
Hemibrain v1.2	5,375	95%	74%	60%
Male CNS v1.0	8,342	97%	75%	62%
BANC 888	4,300	97%	76%	62%

3 Drawing the maps

Each panel shows one calyx in posterior view (Report 02, Table 2): every calyx synapse of the chosen kind is a dot at its position, colored by $w_o(e)$; synapses without a response are gray. The color scale follows Yang's GH146 heatmap PDF: the red–blue map `RdBu_r` with a two-slope normalization, so 0 is white, $v_{\min}$ is the darkest blue and $v_{\max}$ the darkest red:

$$c(v) \;=\; \mathrm{RdBu_r}\big(t(v)\big), \qquad t(v) = \begin{cases} \tfrac{1}{2} + \tfrac{1}{2}\, v / v_{\max} & v \geq 0 \\ \tfrac{1}{2} - \tfrac{1}{2}\, v / v_{\min} & v < 0 \end{cases} \tag{3}$$

$t \in [0, 1]$ is the position along the color map. One scale is used for all odors of a map, set by the whole mean matrix.

The first maps (PA, PAA, MS with the ORN table) drew negative responses as 0; all later maps use the signed two-slope scale above. Within a panel, dots are drawn in order of increasing $|w_o|$, so strong responses sit on top.

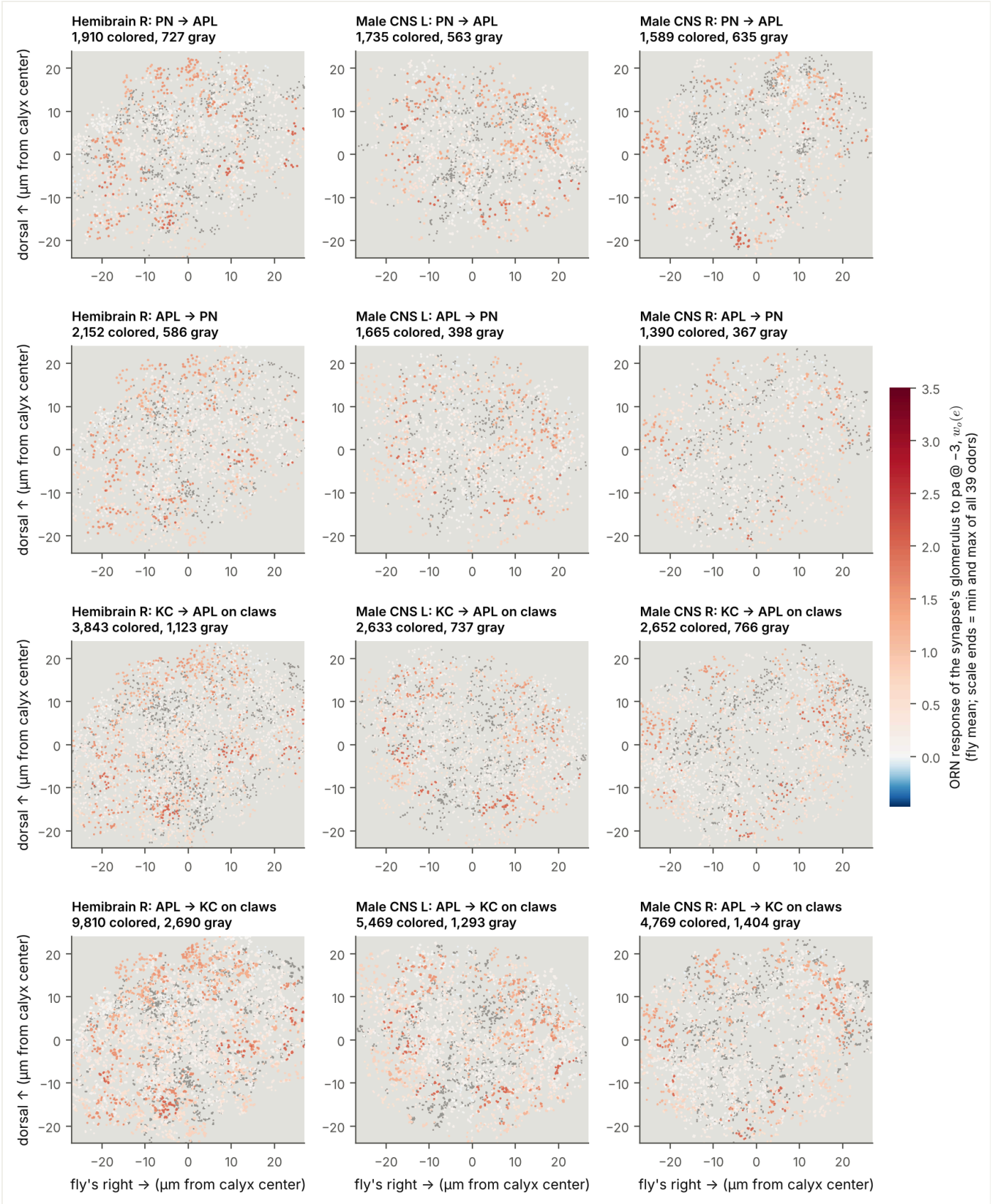


Figure 2. One odor drawn on APL's calyx synapses. Odor = pentyl acetate at 10^{-3} (pa @ −3), ORN fly means. Columns: the three calyces used by the model; rows: the four kinds of APL synapse (PN → APL and APL → PN: all calyx synapses of uniglomerular PNs; KC → APL and APL → KC: only synapses on a claw). Each dot is one synapse in posterior view, in µm from the calyx center (center = midpoint of the 0.2–99.8% range of PN ↔ APL positions). Color = response of the synapse's glomerulus (Equation 2) on the two-slope scale of Equation 3, with ends $v_{\min}$ = −0.48 and $v_{\max}$ = 3.50 (the whole 40-glomerulus × 39-odor ORN matrix); gray = glomerulus not imaged, or no glomerulus. Panel background is light gray so that white (0) stays visible. The delivered maps (`odor_weighted/ORN_per_odor_map_*.png` and the GH146 ones) show every odor and all APLs of all four datasets in this layout.

The strongest boutons and claws of an odor are scattered over the whole calyx rather than gathered in one place: for PA, PAA and MS the response-weighted center of mass lies within 3 µm of the calyx center in FlyWire, hemibrain and male CNS. Some glomeruli do sit in typical places (DM1, the strongest PAA glomerulus, on the ventral rim; DL1, the strongest MS glomerulus, near the center).

Per-glomerulus maps

Two more map sets show, for each of 58 glomeruli in turn, that glomerulus's synapses in color and all others in gray: `glomerulus_map_{PN_to_APL,APL_to_PN}_posterior_view.png` (PN boutons) and `claw_glomerulus_map_{KC_to_APL,APL_to_KC}_posterior_view.png` (claws). They use no odor data.

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4 Odor-pair similarity of on-claw APL synapses

If APL's synapses on claws favored some glomeruli, two odors could look more or less alike "through APL's eyes" than through the glomeruli alone. To test this, each odor becomes a vector over all on-claw KC ↔ APL synapses of a dataset (both APLs), with entry $w_o(e)$; synapses whose glomerulus has no response are left out. The similarity of two odors is the Pearson correlation of their vectors. Because all synapses of a glomerulus carry the same value, this equals a correlation over glomeruli in which glomerulus g counts n_g times, n_g being its number of on-claw synapses:

$$\rho_{ab} = \frac{\sum_g n_g \,(\bar{r}_{g,a} - \mu_a)(\bar{r}_{g,b} - \mu_b)}{\sqrt{\sum_g n_g \,(\bar{r}_{g,a} - \mu_a)^2}\,\sqrt{\sum_g n_g \,(\bar{r}_{g,b} - \mu_b)^2}}, \qquad \mu_a = \frac{\sum_g n_g \bar{r}_{g,a}}{\sum_g n_g} \tag{4}$$

The "responses only" reference sets every $n_g = 1$.

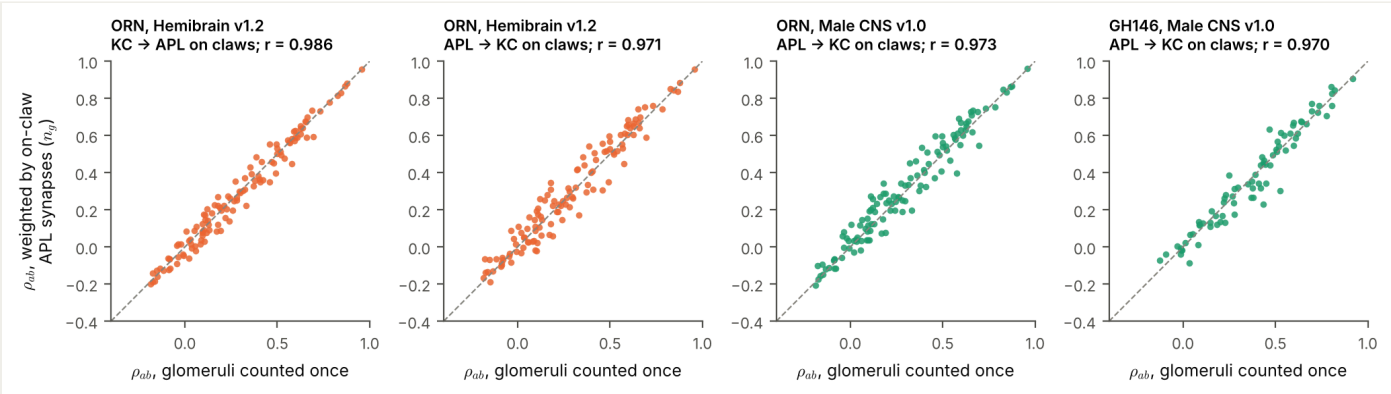


Figure 3. Weighting by APL's on-claw synapses barely changes odor similarity. Each dot is one pair of odors (ORN: 120 pairs of the 16 odors; GH146: 78 pairs of 13). x: Pearson correlation of the two odors' glomerular responses with every glomerulus counted once; y: the same with glomerulus g counted n_g times, n_g = its number of on-claw KC → APL or APL → KC synapses (Equation 4). Dashed line: equal. r in each title = correlation between x and y over pairs. Four of the 16 dataset × direction × response combinations are shown; all are in `odor_weighted/odor_similarity_KC_APL_on_claws.csv`.

In every dataset, direction and response table the weighted similarities track the unweighted ones closely (r 0.945–0.986 over the 16 combinations), and the four connectomes agree with each other (r ≥ 0.947 between any two). So the glomerular distribution of APL's on-claw synapses does not by itself reshape odor similarity. Whether APL's **activity** reshapes the KC representation is a different question, taken up with a model from Report 05 on.

Pearson here, cosine later

This section uses the Pearson correlation (built 2026-10-14). From 2026-10-21 on, Yang asked for cosine similarity, as in their RDMs; the model and the TNT comparison (Reports 05–11) use cosine. `odor_similarity.py` was not switched.

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